

Introduction to Green's Functions

Motivation

Many PDE problems have the form

$$Lu = f$$

where

- ▶ L is a linear differential operator
- ▶ $f(x)$ is a forcing term

Examples:

$$-u'' = f(x)$$

$$-\Delta u = f(x)$$

$$u_t - ku_{xx} = f(x, t)$$

Goal: construct the solution u from the forcing f .

Idea: Response to a Point Source

Suppose the forcing is concentrated at a single point:

$$f(x) = \delta(x - \xi)$$

where δ is the Dirac delta.

We solve

$$LG(x, \xi) = \delta(x - \xi)$$

The function

$$G(x, \xi)$$

is called the **Green's function**.

Green's Function

Let L be a linear differential operator.

A **Green's function** $G(x, \xi)$ satisfies

$$LG(x, \xi) = \delta(x - \xi)$$

together with the same boundary conditions as the original problem.

Interpretation:

- ▶ $G(x, \xi)$ is the response at x
- ▶ produced by a unit source at ξ

Building General Solutions

Suppose

$$Lu = f(x)$$

and we know the Green's function.

Using linearity,

$$f(x) = \int f(\xi) \delta(x - \xi) d\xi$$

Superposition gives

$$u(x) = \int G(x, \xi) f(\xi) d\xi.$$

Thus the solution is an integral operator applied to f .

Example: Poisson Equation

Consider

$$-u''(x) = f(x), \quad 0 < x < L$$

with boundary conditions

$$u(0) = u(L) = 0.$$

The Green's function satisfies

$$-G''(x, \xi) = \delta(x - \xi).$$

Structure of the Green's Function

Away from $x = \xi$ we have

$$G''(x, \xi) = 0.$$

Thus

$$G(x, \xi)$$

is piecewise linear:

$$G(x, \xi) = \begin{cases} A_1x + B_1 & x < \xi \\ A_2x + B_2 & x > \xi \end{cases}$$

Conditions determine the constants.

Jump Condition

Integrate the equation across $x = \xi$:

$$-\int_{\xi-\varepsilon}^{\xi+\varepsilon} G''(x, \xi) dx = \int_{\xi-\varepsilon}^{\xi+\varepsilon} \delta(x - \xi) dx.$$

Taking the limit gives

$$G'_+(\xi) - G'_-(\xi) = -1.$$

Thus the derivative has a jump determined by the delta function.

Resulting Green's Function

The resulting Green's function is

$$G(x, \xi) = \begin{cases} \frac{x(L - \xi)}{L}, & x < \xi \\ \frac{\xi(L - x)}{L}, & x > \xi \end{cases}$$

The solution becomes

$$u(x) = \int_0^L G(x, \xi) f(\xi) d\xi.$$

Heat Equation Green's Function

For the heat equation on $\mathbb{R}$

$$u_t = ku_{xx}$$

the Green's function is

$$G(x, t) = \frac{1}{\sqrt{4\pi kt}} \exp\left(-\frac{x^2}{4kt}\right).$$

This represents heat spreading from a point source.

Heat Equation Solution

If

$$u_t = ku_{xx}, \quad u(x, 0) = f(x)$$

then

$$u(x, t) = \int_{-\infty}^{\infty} G(x - \xi, t) f(\xi) d\xi.$$

Thus the solution is a convolution with the heat kernel.

Interpretation

Green's functions provide

- ▶ the response to a point source
- ▶ an integral representation of solutions
- ▶ a bridge between PDEs and integral operators

They play a central role in

- ▶ potential theory
- ▶ heat flow
- ▶ wave propagation
- ▶ quantum mechanics